

Final Report

Group 21

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1 Motivation

ScrimFinder was born from the need for a more structured and data-driven approach to competitive "scrims" (practice matches) in League of Legends. While the official game offers matchmaking, it lacks the depth required for teams and individual players to organize balanced practice sessions, track progress through custom-built ranking systems (MMR), and gain actionable insights through post-game ML-driven analysis. By leveraging a massive 78GB dataset of EUW matches, ScrimFinder provides a platform that not only matches players based on skill but also analyzes their performance to offer personalized coaching tips. Our goal was to build a truly cloud-native application that demonstrates high availability, scalability, and advanced data processing capabilities.

2 Dataset Characterization

- **Topic:** League of Legends Matches
 - **URL:** Kaggle - Matches Raw EUW
 - **Size:** Approximately 78 GB
 - **Release Date:** January 2026
-

3 Business Capabilities

3.1 Competitive Ranking

Facilitates real-time visualization of global and local leaderboards. Includes simplified performance metrics such as Damage dealt and KDA ratios.

3.2 Custom Match Management

Enables the creation of balanced custom matches. Matchmaking is handled via a global/local queue system integrated with the project's ranking logic.

3.3 Custom Match History

Provides a centralized repository for past custom match statistics. Includes robust functionality for custom filtering and sorting based on specific user requirements.

3.4 Team Draft and Build Analyzer

Offers personalized post-game analysis. Players utilizing the API receive automated insights regarding gameplay performance and actionable tips for improvement.

4 Use Cases

4.1 Match & Ranking Management

Use Case	Student Name
Leaderboards	Diogo Sousa
User Ranking	Diogo Sousa
Match Finding	Diogo Sousa
Create / Cancel Match	Diogo Sousa

Table 1: Match & Ranking Management Use Cases per Student

4.2 Data Visualization & Analysis

Use Case	Student Name
Match History	Bruno Faustino
Advanced History Filtering / Sorting / Paging	Bruno Faustino
Automatic Detail Filling (by interacting with RIOT's API)	Bruno Faustino
Draft Analysis	Rodrigo Neto
Build Analysis	Rodrigo Neto
Player Performance Analysis	Rodrigo Neto

Table 2: Data Visualization & Analysis Use Cases per Student

5 API Specification

5.1 Player Management - Diogo Sousa

Operations for registering players and linking their Discord accounts with external League of Legends identities.

Verb	Endpoint	Description
POST	/api/v1/matchmaking/ players	Register a new player in the platform.
GET	/api/v1/matchmaking/ players/{id}	Get internal player details.
GET	/api/v1/matchmaking/ players/{id}/lobbies	Get all lobbies associated with a player.
GET	/api/v1/matchmaking/ players/{id}/tickets	Get all active tickets for a player.
DELETE	/api/v1/ranking/players/ links/{gameName}/ {tagLine}	Unlink a League of Legends account.
GET	/api/v1/ranking/players/ {playerId}	Get player profile and MMR.
PUT	/api/v1/ranking/ players/{playerId}/ link-lol-account	Link a Riot account to a player.
GET	/api/v1/ranking/ players/{playerId}/ primary-account	Get the primary linked Riot account.
PUT	/api/v1/ranking/ players/{playerId}/ set-primary-account	Set the primary Riot account.
POST	/api/v1/ranking/players/ {playerId}/sync-mm	Sync MMR from external services.

Table 3: Player Management REST API

5.2 Queue Management - Diogo Sousa

Operations for managing matchmaking queues, defining rules, and setting MMR requirements.

Verb	Endpoint	Description
POST	/api/v1/matchmaking/queues	Create a new global or local queue.
GET	/api/v1/matchmaking/queues/{id}	Get queue configuration and status.
PUT	/api/v1/ranking/queue/{queueId}	Update queue MMR rules.
GET	/api/v1/ranking/queue/{queueId}	Get queue details from Ranking.
DELETE	/api/v1/ranking/queue/{queueId}	Delete a queue.

Table 4: Queue Management REST API

5.3 Matchmaking - Diogo Sousa

Core logic for joining queues, accepting match proposals, and completing matches.

Verb	Endpoint	Description
POST	/api/v1/matchmaking/tickets	Join a matchmaking queue.
POST	/api/v1/matchmaking/tickets/matches/{matchId}/complete	Finalize match and report results.
PUT	/api/v1/matchmaking/tickets/matches/{matchId}/link	Link external Game ID to match.
POST	/api/v1/matchmaking/tickets/matches/{matchId}/players/{playerId}/accept	Accept a match proposal.
POST	/api/v1/matchmaking/tickets/matches/{matchId}/players/{playerId}/decline	Decline a match proposal.
DELETE	/api/v1/matchmaking/tickets/{ticketId}	Leave a matchmaking queue.
GET	/api/v1/matchmaking/tickets/{ticketId}/lobby	Get current lobby for a ticket.

Table 5: Matchmaking REST API

5.4 Leaderboard - Diogo Sousa

Retrieval of global and queue-specific rankings.

Verb	Endpoint	Description
GET	/api/v1/ranking/leaderboards	Get global leaderboard (paginated).
GET	/api/v1/ranking/leaderboards/queues/{queueId}	Get queue-specific leaderboard.

Table 6: Leaderboard REST API

5.5 Player Ranking - Diogo Sousa

Operations to track individual player MMR and ranking history across different queues.

Verb	Endpoint	Description
GET	/api/v1/ranking/players/{playerId}/queue-rankings	Get all queue rankings for a player.
GET	/api/v1/ranking/players/{playerId}/queue-rankings/{queueId}	Get ranking for a specific queue.

Table 7: Player Ranking REST API

5.6 Match History - Bruno Faustino

Historical repository of match statistics with advanced filtering and paging.

Verb	Endpoint	Description
GET	/api/v1/history/matches	Get paginated match history with filters.
GET	/api/v1/history/matches/{riotMatchId}	Get detailed stats for a single match.

Table 8: Match History REST API

5.7 Match Filling - Bruno Faustino

Interaction with Riot API to enrich match data.

Verb	Endpoint	Description
GET	/api/v1/riot/matches/{matchId}	Get simplified stats from Riot API.
GET	/api/v1/riot/matches/{matchId}/raw	Get raw match data for analysis.

Table 9: Match Filling REST API

5.8 Player Filling - Bruno Faustino

Retrieval of external player accounts and stats from Riot.

Verb	Endpoint	Description
GET	/api/v1/riot/players/{server}/{name}/{tag}	Get Riot account details by summoner name.

Table 10: Player Filling REST API

5.9 System - Rodrigo Neto

Health checks and monitoring endpoints for analysis and training services.

Verb	Endpoint	Description
GET	/api/v1/analysis/	Service root.
GET	/api/v1/analysis/health	General health status.
GET	/api/v1/analysis/q/health/ready	Readiness probe.
GET	/api/v1/analysis/q/health/live	Liveness probe.
GET	/api/v1/training/	Service root.
GET	/api/v1/training/health	General health status.
GET	/api/v1/training/q/health/ready	Readiness probe.
GET	/api/v1/training/q/health/live	Liveness probe.

Table 11: System REST API

5.10 Training - Rodrigo Neto

Management of ML training jobs for performance and draft models.

Verb	Endpoint	Description
POST	/api/v1/training/jobs	Start a new ML training job.
GET	/api/v1/training/jobs	List all training jobs.
GET	/api/v1/training/jobs/{job_id}	Get job status and metrics.
POST	/api/v1/training/jobs/{job_id}/cancel	Stop a running job.
POST	/api/v1/training/jobs/{job_id}/deploy	Deploy the trained model.

Table 12: Training REST API

5.11 Datasets - Rodrigo Neto

Creation and curation of training datasets from ingested games.

Verb	Endpoint	Description
POST	/api/v1/training/datasets	Register a new dataset definition.
GET	/api/v1/training/datasets	List curated datasets.
POST	/api/v1/training/datasets/build	Create and build a dataset from games.
GET	/api/v1/training/datasets/{dataset_id}	Get dataset metadata.
DELETE	/api/v1/training/datasets/{dataset_id}	Delete a dataset.
POST	/api/v1/training/datasets/{dataset_id}/build	Rebuild an existing dataset.

Table 13: Datasets REST API

5.12 Games - Rodrigo Neto

Ingestion and retrieval of match data for training purposes.

Verb	Endpoint	Description
POST	/api/v1/training/games	Ingest a single match.
GET	/api/v1/training/games	List all ingested games.
POST	/api/v1/training/games/batch	Ingest matches in batches (up to 1000).
GET	/api/v1/training/games/{game_id}	Fetch stored match details.
DELETE	/api/v1/training/games/{game_id}	Delete an ingested game.

Table 14: Games REST API

5.13 Models - Rodrigo Neto

Registry for versioned ML models and their active status.

Verb	Endpoint	Description
GET	/api/v1/training/models	List all registered models.
GET	/api/v1/training/models/active	List currently active models.
GET	/api/v1/training/models/{model_id}	Get model performance metrics.
DELETE	/api/v1/training/models/{model_id}	Remove a model from registry.
POST	/api/v1/training/models/{model_id}/activate	Set model as active for analysis.
POST	/api/v1/training/models/{model_id}/deactivate	Deactivate a model.

Table 15: Models REST API

5.14 Features - Rodrigo Neto

Extraction of structured feature vectors from raw match data.

Verb	Endpoint	Description
POST	/api/v1/training/features/extract	Extract features from match data.
GET	/api/v1/training/features/{game_id}	Get cached feature vectors.
DELETE	/api/v1/training/features/{game_id}	Delete cached features.

Table 16: Features REST API

5.15 Analysis - Rodrigo Neto

Predictive analysis for drafts, builds, and player performance.

Verb	Endpoint	Description
POST	/api/v1/analysis/draft	Predict team win probability.
POST	/api/v1/analysis/build	Score item build effectiveness.
POST	/api/v1/analysis/player	Aggregated performance benchmarking.
POST	/api/v1/analysis/game	Detailed post-game breakdown.
POST	/api/v1/analysis/champion	Global champion stats from EUW dataset.

Table 17: Analysis REST API

6 Functional Requirements

6.1 Matchmaking - Diogo Sousa

- Management of the matchmaking queue lifecycle, including ticket creation, status tracking, and cancellation;
- "On Join" matchmaking logic to evaluate and form lobbies immediately upon player entry;
- Implementation of a Standard Queue for 10-player lobby formation;
- Implementation of a Role Queue ensuring team balance across all five roles (TOP, JUNGLE, MID, BOT, SUPP);
- Lobby lifecycle control, including match acceptance/rejection and automatic lobby dissolution on failure;
- Coordination of match states from initial discovery to finalization and result reporting.

6.2 Ranking - Diogo Sousa

- Real-time calculation and updates of player skill ratings (ELO/MMR) based on match performance;
- Management of global and queue-specific leaderboards with paginated retrieval;
- Automatic player profile initialization and rank mapping via Riot account integration;
- Support for multiple ranking systems tailored to specific queues.

6.3 Match History - Bruno Faustino

- Local store of simplified match information relevant to the user (winning team, rank, roles of each player, number of kills, deaths, assists, healing, damage,..., per match, per team and per player), that can be inserted either manually, with validation, or by providing the match's ID in Riot's API;
- Match filtering based on any criteria the user defines from the simplified information, including obtaining all matches of a given player, all matches where the player had the most kills, ...;
- Match sorting based on any criteria;
- Match retrieval with paging to avoid fetching enormous amounts of data all at once;
- Soft deletion of matches from the history;
- New match data forwarding to the data analysers.

6.4 Detail Filling - Bruno Faustino

- Direct contact with Riot's API to obtain match and player information;
- Automatic filtering of information to focus more on the user's needs;
- Retrieval of the full raw data to be used for data analysis;
- Retrieval of player information (rank, wins, losses, points, ...).

6.5 Training - Rodrigo Neto

- Ingestion of raw match data in batches with automatic deduplication and metadata storage;
- Extraction and versioning of structured feature vectors for different analytical models;
- Creation and management of curated, optimized training datasets with configurable filtering;
- Asynchronous execution of ML training jobs for draft, build, and performance models;
- Real-time monitoring of training progress and lifecycle management of active jobs;
- Maintenance of a versioned model registry with performance metrics.

6.6 Data Analysis - Rodrigo Neto

- Win probability prediction for team compositions;

- Identification of team synergies, counter-matchups, and win conditions;
 - Evaluation of item build effectiveness and possibly suggestion of alternatives;
 - Automated benchmarking of player performance metrics against global/rank-tier averages;
 - Global champion performance tracking including win, pick, and ban rate analytics.
-

7 Non-Functional Requirements

In accordance with the project guidelines, we implemented several non-functional improvements to ensure the system is robust, scalable, and secure.

7.1 Microservices Architecture - Diogo Sousa

The core architecture focuses on high availability and loose coupling through message-driven patterns.

- **Database per Service:** Each microservice (Matchmaking, Ranking, History) maintains its own isolated PostgreSQL database, ensuring data encapsulation and independent scaling.
- **Message-Driven (Pub/Sub):** Utilization of **RabbitMQ** for asynchronous, decoupled communication (e.g., Matchmaking publishing game results that are consumed by History and Training services).

7.2 Serverless & FaaS Integration - Bruno Faustino

To optimize resource usage, specific background tasks (such as Detail Filling via Riot's API) were designed to transition to a Serverless model.

- **Implementation:** Evaluated **Quarkus Funqy** for a portable, provider-agnostic function-as-a-service (FaaS) experience to handle bursty API polling.

7.3 Data Science & Machine Learning - Rodrigo Neto

- **Batch ML Pipelines:** Engineered a dedicated Training service using **scikit-learn** and **LightGBM** to process the 78GB dataset in batches, extracting features and training predictive models without impacting real-time API performance.

7.4 DevOps & Automation - Bruno Faustino & Rodrigo Neto

- **GitOps:** **Argo CD** is used to automatically synchronize the Kubernetes cluster state with the Git repository.
 - **Infrastructure as Code (IaC):** Provisioning Google Kubernetes Engine (GKE) and secrets management using **Terraform**.
-

8 Application Architecture

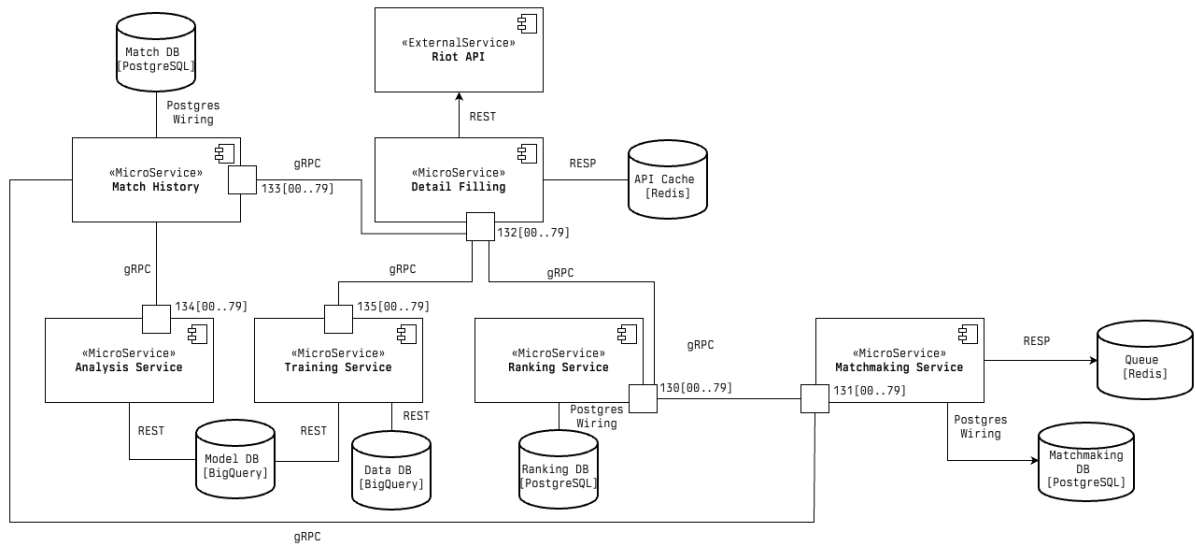


Figure 1: Application Architecture

8.1 Matchmaking Service - Diogo Sousa

- **Database: Redis (Queue/Locks) + PostgreSQL (Lobby Persistence)**
- **Usage:**
 - **Redis:** Manages the active matchmaking tickets and distributed locks for thread-safe lobby formation.
 - **PostgreSQL:** Stores Lobby, Match, and MatchTicket entities to track the lifecycle of a match.

8.2 Ranking Service - Diogo Sousa

- **Database: PostgreSQL (Ranking DB)**
- **Usage:** Persists long-term player profiles, Discord/Riot account linkages, and MMR history.

8.3 Match History Service - Bruno Faustino

- **Database: PostgreSQL (History DB)**
- **Usage:** Indexes detailed match statistics for players and teams, supporting complex filtering.

8.4 Detail Filling Service - Bruno Faustino

- **Database: Redis (API Cache)**
- **Usage:** Caches Riot API responses for accounts and match details to minimize external calls.

8.5 Training Service - Rodrigo Neto

- **Database: BigQuery (Model DB) / Persistent Volume**
- **Usage:** Responsible for training and storing machine learning models using historical match data.

8.6 Analysis Service - Rodrigo Neto

- **Usage:** Consumes models to provide real-time and post-game insights (draft analysis, performance metrics).

9 Technical Architecture

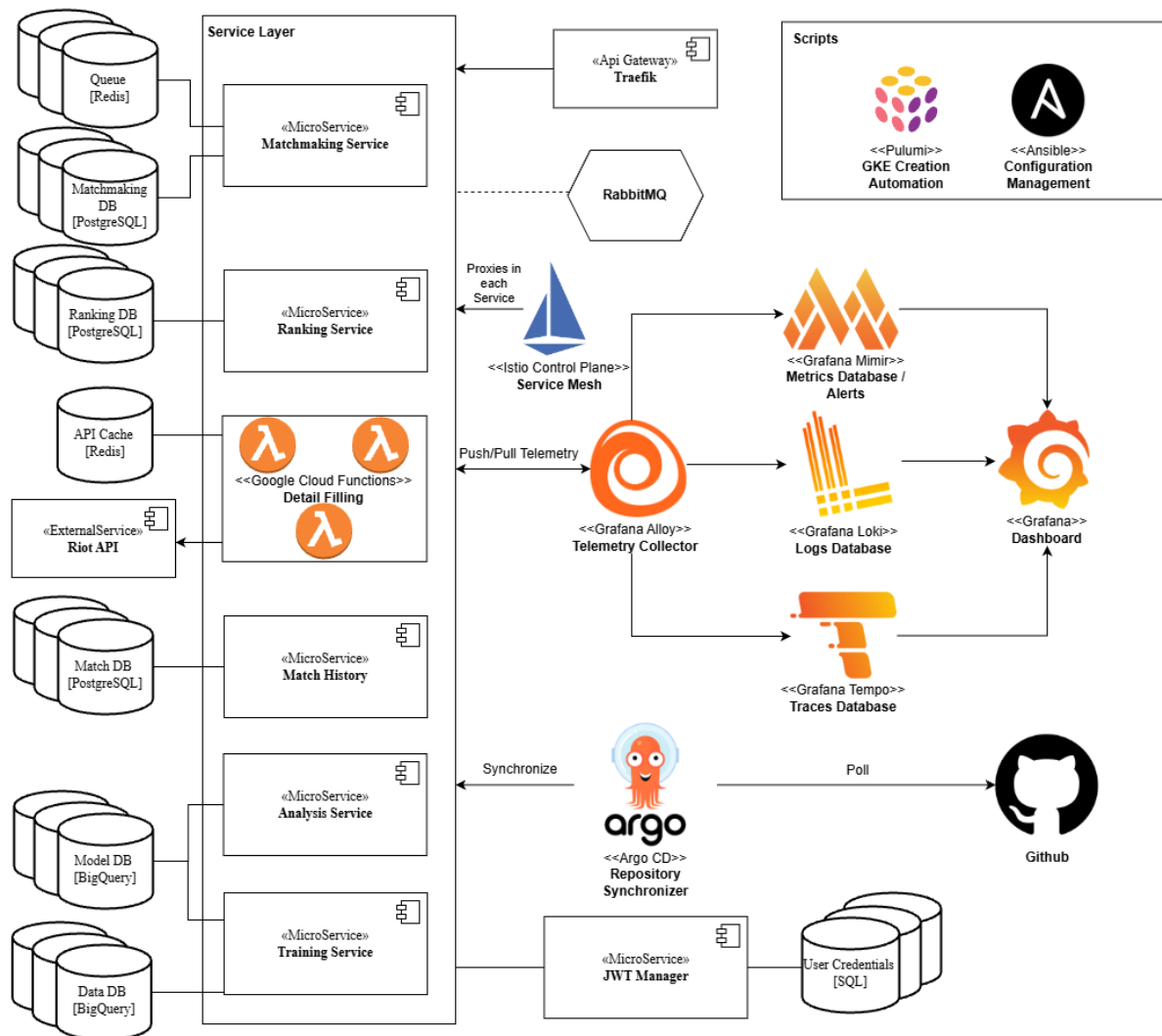


Figure 2: Technical Architecture

10 Implementation (High Level Description)

The system is implemented as a suite of polyglot microservices:

- **Core Backend (Java/Quarkus):** The Matchmaking, Ranking, History, and Detail Filling services are built using Java 21 and the Quarkus framework. This choice provides excellent performance, low memory footprint (ideal for Kubernetes), and native support for gRPC, REST (RESTEasy), and Hibernate ORM for PostgreSQL interaction.
 - **Data Science Backend (Python/FastAPI):** The Training and Analysis services are built in Python using FastAPI. This allows us to leverage the extensive Python ML ecosystem (Pandas, scikit-learn, LightGBM) while maintaining high-performance asynchronous API endpoints.
 - **Communication:** Synchronous inter-service communication is handled via strict **gRPC** contracts (compiled for both Java and Python), while asynchronous events (like match completions) are broadcasted via **RabbitMQ** exchanges.
-

11 Deployment Scripts

11.1 Local Checks

```
./scripts/format.sh
./scripts/test.sh
```

Python gRPC files are generated, not committed:

```
./scripts/python/generate-grpc.sh
```

11.2 Deployment

Required environment:

```
export SCRIM_PROJECT_ID="your-gcp-project"
export SCRIM_REGION="europe-west1"
export SCRIM_REPO_NAME="scrimfinder"
export SCRIM_CLUSTER_NAME="scrimfinder"
export RIOT_API_KEY="RGAPI-..."
export SCRIM_DB_USER="scrim_admin"
export SCRIM_DB_PASSWORD="secure_password"
```

Deploy infrastructure, images, Cloud Functions, and the Helm application:

```
./scripts/deploy.sh
```

Tear down the stack:

```
./scripts/shutdown.sh
```

12 Tuning

System performance is optimized through precise resource allocation and automatic scaling policies.

12.1 Resource Requests and Limits

Each service is configured with baseline requests and hard limits to ensure fair scheduling and prevent OOM (Out of Memory) events:

- **Java Services:** Typically 400m-500m CPU and 512Mi Memory.
- **Training Service:** High resource allocation (600m CPU, 2Gi Memory) to handle ML workloads.
- **Analysis Service:** 500m CPU and 1Gi Memory for real-time inference.

12.2 Scaling Policies

- **Horizontal Pod Autoscaler (HPA):** Enabled for all services, scaling from 1 to 3 replicas based on a 70-80% CPU utilization threshold.
- **Vertical Pod Autoscaler (VPA):** Globally enabled in recommendation mode to provide insights for future resource adjustments without disrupting Argo CD sync.

12.3 Health Probes

Comprehensive health monitoring is implemented using SmallRye Health:

- **Readiness Probes:** Ensure traffic is only routed once the service has established database connections and linked with gRPC peers.
 - **Liveness Probes:** Automatically restart containers that enter a deadlocked state.
 - **Startup Probes:** Configured with extended failure thresholds (up to 60 for Training service) to accommodate longer initialization times during cold starts.
-

13 CI/CD Pipeline

The ScrimFinder automation pipeline is implemented via GitHub Actions, providing an exhaustive validation flow for every contribution.

13.1 Continuous Integration (CI)

- **Fast Tests:** On every push to `main` or `work/**`, a matrix job runs unit and light integration tests for all Java (Quarkus) and Python (FastAPI) services. This includes linting with `spotless` and `ruff`.
- **Heavy Tests:** Pull requests trigger a complete integration suite.
- **Ephemeral Stacks:** For critical validation, the CI provisions a temporary GKE cluster using Terraform, deploys the entire application stack via Helm, and runs system-wide tests against the live environment before tearing it down.

13.2 Continuous Deployment (CD)

- **GitOps with Argo CD:** The production state is managed in the `k8s/application.yaml` manifest. Argo CD monitors the repository and automatically synchronizes the cluster state whenever a change is merged to the main branch.
-

14 Test, Evaluation Techniques and Results (to target group)

14.1 Documentation Feedback - Diogo Sousa

- **Style Discrepancy:** The query parameter for pagination size is documented as `pageSize` (camelCase) while the response object uses `page_size` (snake_case). The API Gateway correctly accepts the `pageSize` parameter and returns the corresponding schema, but this inconsistency should ideally be unified.
- **Error Response Documentation:** While standard HTTP error codes (400, 401, 404, 500, 502) are consistently mapped across endpoints using reusable components (e.g., `BadRequestText`), they could be more descriptive. The schema indicates they return a `text/plain` string without explaining *what* triggers them (e.g., what specific validation failure causes a 400, or under what conditions a 404 is thrown for a specific endpoint).

14.2 Testing Results - Diogo Sousa

14.2.1 Smoke Testing (Endpoint: [GET/api/listings/search](#))

The smoke test was executed using a **Python script** checking the endpoint. The test validated the HTTP 200 response, JSON validity, and the presence of all required top-level fields (`total`, `page`, `page_size`, `listings`) and listing level attributes.

- **Basic Paginated Search:** Passed. Returned 5 listings with a reported total of 50,000 records. Latency: $\approx 141\text{ms}$.
- **Filtered Search (Toyota Corolla):** Passed. Returned 5 listings with a reported total of 300 records. Latency: $\approx 191\text{ms}$.

No broken links or malfunctioning behaviors were detected.

14.2.2 Stress Testing (Endpoint: [GET/api/listings](#))

The stress test was executed using **Locust** against endpoint.

- **Concurrency Settings:** 100 concurrent users, spawning at a rate of 20 users per second, run duration of 30 seconds.
- **Performance observed:**
 - **Total Requests:** 6,899
 - **Failures:** 0 (0.00%)
 - **Requests Per Second (RPS):** Peaked at ≈ 275 req/s (average 248.81 req/s over the duration).
 - **Latency Statistics:** Average = 72ms, Median = 62ms, 95th percentile = 93ms, 99th percentile = 150ms.

The endpoint proved to be highly stable under the tested load.

14.3 Documentation Feedback - Bruno Faustino

- **Some positives:**
 - The API documentation was in general easy to understand due to the modularity on the groups of endpoints and the descriptive information tagged with each endpoint.
 - The introductory paragraph gives a quick overview of all the use cases of the application and in which modules they can be found.
 - In general, the exceptions portray in which cases the app may fail (although they could be explained in more detail according to the endpoint in some cases).
- **A few suggestions:**
 - For the chat endpoints, a 404 code should be thrown if the chat id does not exist in the database.
 - It would be helpful to have default values already filled-in in the body of the POST methods that serve as an example to further improve the understandability of the API (for some fields it is not trivial what the expected format is, like `fuel_type`, `drive_type` and `end_time`).
 - Although it is well explained that the user needs to add the Host header before calling the post/put/delete endpoints, it is never explained that the user needs to be logged in and that the resulting token needs to be added to the header: a quick mention in either the introductory paragraph or the 401 Unauthorized exception in the documentation would generate less confusion and save a few 401s.

14.4 Testing Results - Bruno Faustino

14.4.1 Smoke Testing (Endpoint: [GET/api/auctions/{auction_id}/bids](#))

The smoke test was executed using **pytest** and **requests** modules of **Python**.

Here is the sequence of API calls as a setup before testing the actual endpoint:

1. **POST**/[api/user/login](#) for the auth token.

2. **POST**/api/listings for listing_id.
3. **POST**/api/auctions for auction_id.
4. 3x **POST**/api/auctions/{auction_id}/bid.

After calling the endpoint: **GET**/api/auctions/{auction_id}/bids, it is expected that the command returns successfully (200) and with 3 auction bids in the created auction for the created listing.

The smoke test passed!

14.4.2 Stress Testing (Endpoint: **GET**/api/market/average_price)

The stress test was executed using **Locust** against the endpoint.

- **Concurrency Settings:** 10000 concurrent users, spawning at a rate of 5 users per second, running until failing.
- **Performance observed:**
 - **Total Requests:** 250
 - **Failures:** 4 (1.60%)
 - **Requests Per Second (RPS):** Very inconsistent, around 19.61 req/s.
 - **Latency Statistics:** Average = 1009ms, Median = 190ms, 95th percentile = 4000ms, 99th percentile = 6700ms.

The system fails a few times at 19.61 req/s due to the unavailability of an upstream service (502).

During other rounds of testing, the system could fail at less or more than that value, sometimes managing to survive without failures until around 50 req/s, so it is inconsistent, and the number of failures oscillates a lot between load tests, up to the point of reaching 99% failures.

The endpoint proved unable to sustain loads above 50 req/s, and shows signs of failing around 19.61 req/s.

14.5 Documentation Feedback - Rodrigo Neto

- **Missing Content-Type Header:** Successful JSON responses do not include the **Content-Type: application/json** header, which can break strict HTTP clients.
- **Poor Overload Behaviour:** Under high load, the backend does not gracefully degrade (e.g., return 429 or 503). Instead, the gateway returns 502 **Bad Gateway** after extremely long timeouts (up to 47 seconds).

14.6 Testing Results - Rodrigo Neto

14.6.1 Smoke Testing (Endpoint: **GET**/api/market/price-comparison)

The smoke test was executed using **Postman** checking the endpoint. The test validated HTTP 200 responses, JSON validity, pagination, grouping, and sorting behaviors.

- **Happy Path:** Passed with warnings. Valid JSON, correct pagination and sorting. **Content-Type** assertion failed (header omitted).

- **Validation:** Partial pass (3/6). Missing parameters returned 400. Invalid enum values returned 502 `Bad Gateway` instead of 400.

Some malfunctioning behaviors were detected during validation tests.

14.6.2 Stress Testing (Endpoint: [GET/api/listings/stats/by-location](#))

The stress test was executed using **Locust** against the endpoint.

- **Concurrency Settings:** Step-load ramping from 10 to 200 concurrent users.
- **Performance observed:**
 - **Total Requests:** 3,524
 - **Failures:** 123 (3.49%)
 - **Requests Per Second (RPS):** Peaked at ≈ 50 req/s.
 - **Latency Statistics:** Average = 7,280ms, Median = 460ms, 95th percentile = 34,000ms, 99th percentile = 38,000ms.

The endpoint proved unable to sustain loads above ≈ 50 RPS, leaving the upstream backend in an unhealthy state.

15 Corrections Implemented after Feedback from Testing

Following the "Cloudy Day" testing phase and internal smoke tests, several critical improvements were implemented to enhance system reliability and client experience.

15.1 Improved Error Handling in Player Registration

Initial feedback indicated that the `POST /api/v1/matchmaking/players` endpoint returned a `500 Internal Server Error` when a duplicate UUID was provided, whereas a duplicate Discord username correctly returned a `409 Conflict`. We updated the `PlayerServiceImpl` to perform an explicit check for existing IDs before persistence. This ensures that all conflict scenarios (ID or Username) return a clear `409` status with a specific error message, improving the API's predictability.

15.2 Standardized Exception Mapping

We refined the `GlobalExceptionHandler` to move away from generic `REMOTE_SERVICE_ERROR` codes for client-side issues. HTTP status codes are now accurately reflected in the error response payload, using `BAD_REQUEST` or `NOT_FOUND` where appropriate. This directly addresses feedback regarding the ambiguity of error messages under high load and validation failures.

15.3 Documentation Consistency

OpenAPI annotations were updated to explicitly state uniqueness constraints and provide example payloads for complex enums, reducing the "trial and error" approach previously required by external testers.

16 Cost Analysis

The following table outlines the estimated monthly operational costs for ScrimFinder on Google Cloud Platform (Region: **europe-west1**), calculated using the Official GCP Pricing Calculator. In this architecture, all databases (PostgreSQL, Redis) and message brokers (RabbitMQ) are self-hosted directly on the GKE cluster using persistent storage, rather than managed cloud services. Costs are estimated in Euros (€).

Component	Details	Est. Cost (EUR)
GKE Cluster ¹	3-5 nodes (e2-standard-4 Spot) + Management fee	€170.00
Persistent Storage ²	SSD Storage for PostgreSQL, Redis, and AI data	€145.00
BigQuery ³	78GB active storage + Analysis slots	€28.00
Other ⁴	Artifact Registry, Cloud Functions, Traffic	€14.00
Total		€357.00

Table 18: Estimated Monthly Infrastructure Cost

16.1 FinOps and Optimization Measures

To keep the platform cost-effective, we implemented the following strategies:

- **BigQuery for ML:** Offloading massive match history storage to BigQuery instead of relational SQL databases significantly reduces storage costs.
- **Redis Caching:** Aggressively caching Riot API responses reduces the need for expensive high-concurrency calls and helps stay within lower API tier limits.
- **Autoscaling:** HPA ensures we only run multiple replicas during peak hours, while VPA helps identify over-provisioned services that can be downsized.

17 Conclusions

TODO

¹GKE Pricing: Scaled to handle both application logic and stateful database workloads.

²Persistent Disk Pricing: Based on regional SSD (pd-ssd) for high-IOPS database performance.

³BigQuery Pricing: Scaled for the training service’s feature extraction workloads.

⁴Includes external load balancer and public IP addresses.